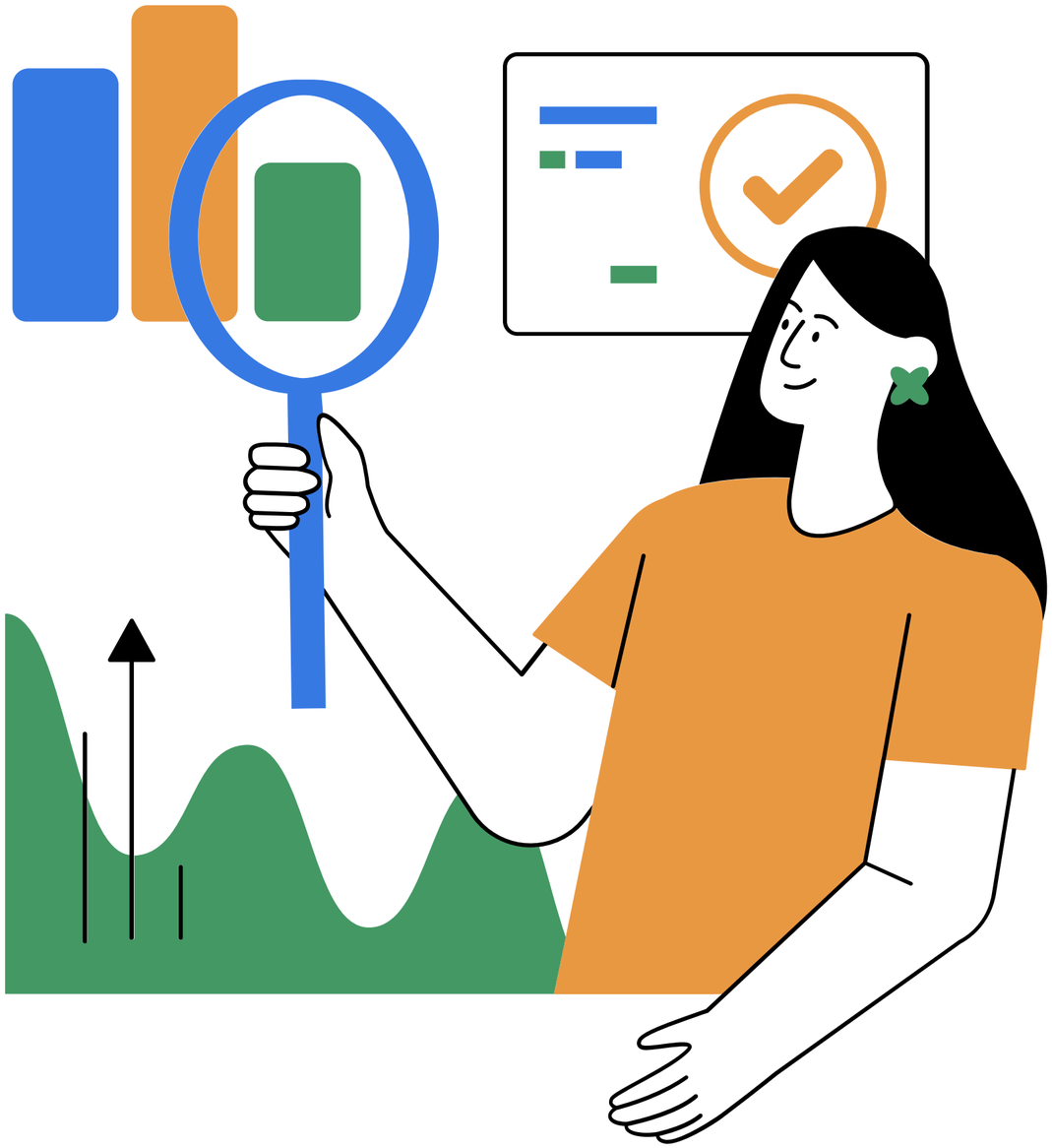
Spotify Virality Study



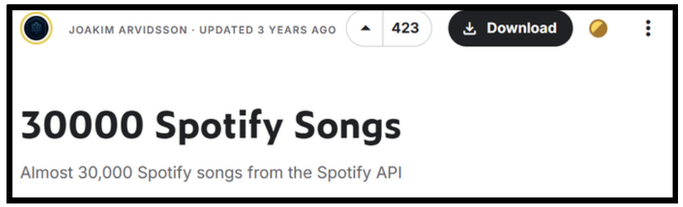
The Question

Can we predict how popular a song will be based purely on its audio DNA?

- **32,833** tracks from the Spotify Web API
- **23** features: danceability, energy, loudness, tempo, valence, and more
- Target: track_popularity (0–100 continuous score)

The Data

Source: Kaggle



32,833 tracks, **23** features

Features:

AUDIO
- danceability
- energy
- acousticness
- liveness
- valence
- loudness
- speechiness
- instrumentalness
- tempo

CATEGORICAL
- playlist_genre
- playlist_subgenre

TEMPORAL
- release_year
- release_month

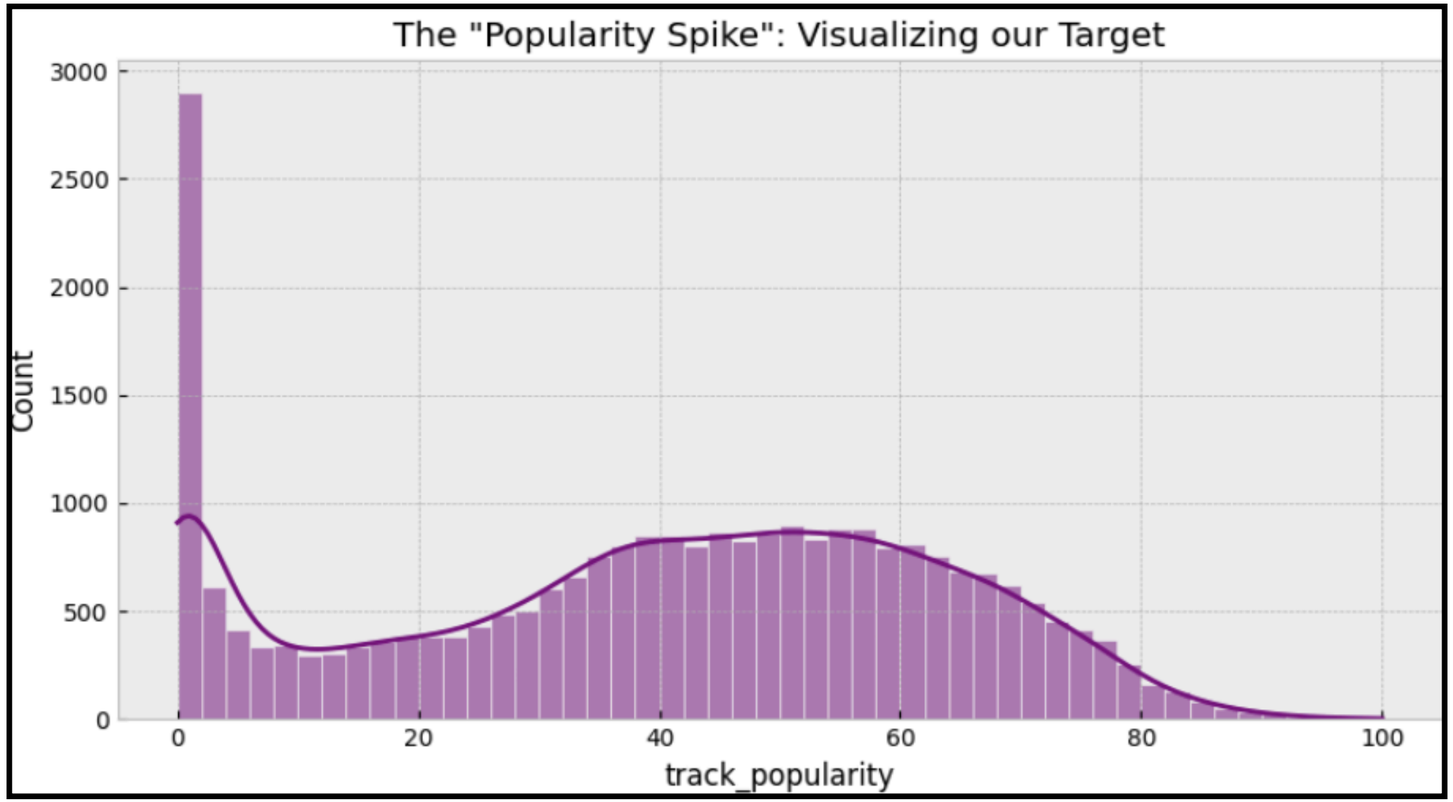
ENGINEERED
- intensity_index
- dance_tempo_ratio

Cleaning the Data

Step	Result
Remove duplicate track IDS	**-3961** entries
Dropped rows with nulls	**-315** entries
Final dataset	**28,557** unique tracks

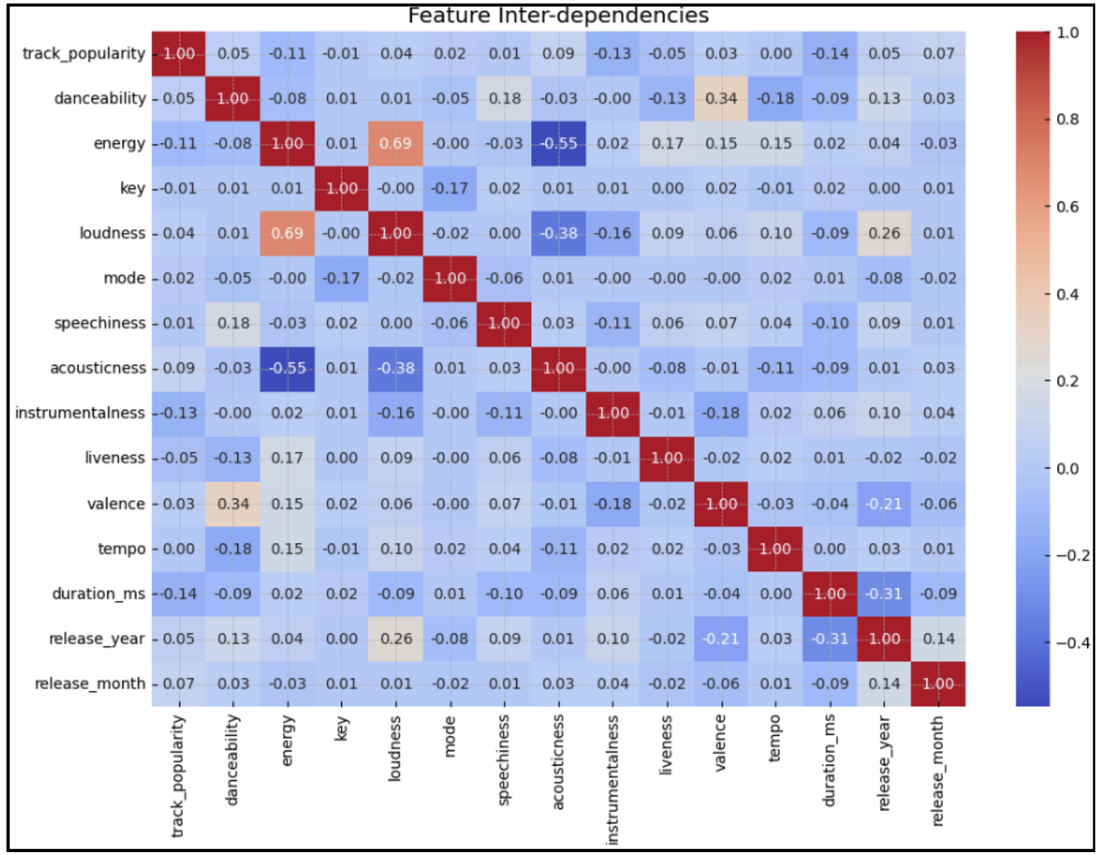
- 800 loudness outliers detected via IQR - retained, handled with RobustScaler
- Release dates converted to datetime → extracted year and month as features
- No imputation needed - dataset was clean post-deduplication

Visualising Data



- Popularity is heavily right-skewed - most tracks cluster near zero
- Only a small fraction of tracks achieve scores above 70
- This reflects the reality of streaming: **a few tracks capture most attention**
- This skew would later make "hit prediction" a difficult classification problem

Do Audio Features Predict Popularity?



Barely.

- Highest correlation with popularity: instrumentalness at -0.13
- danceability, energy, valence, tempo - all under 0.11
- No single audio feature is a meaningful linear predictor

However, the heatmap does show:

- **energy ↔ loudness**: 0.69 - these features move together
- **acousticness ↔ energy**: -0.55 - acoustic tracks are distinctly quieter and calmer
- **valence ↔ danceability**: 0.34 -happier songs tend to be more danceable

Modeling Approach

We built a unified preprocessing pipeline and tested five models in sequence:

RobustScaler + OneHotEncoder → Model

Stage	Model	Purpose
Baseline	Linear Regression	Establish floor
Baseline	Ridge Regression	Check for overfitting
Non-linear	Polynomial Ridge (degree 2)	Capture curves
Ensemble	Random Forest	Bagging
Ensemble	XGBoost	Gradient boosting
Final	Stacking Ensemble	Combine best models

*** Categorical features (genre, subgenre) encoded via OneHotEncoder and fed into every model.**

Results

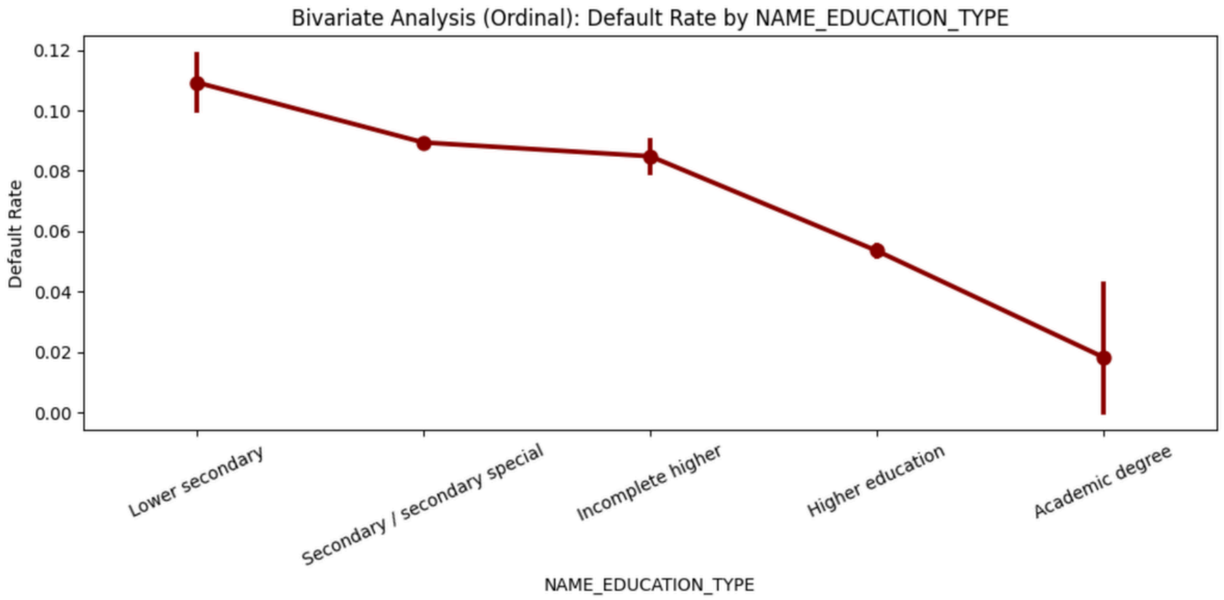
	Model	RMSE
4	XGBoost	22.351500
2	Polynomial Ridge (Degree 2)	22.599800
3	Random Forest	22.724000
1	Ridge Regression	22.995200
0	Linear Regression	22.995300

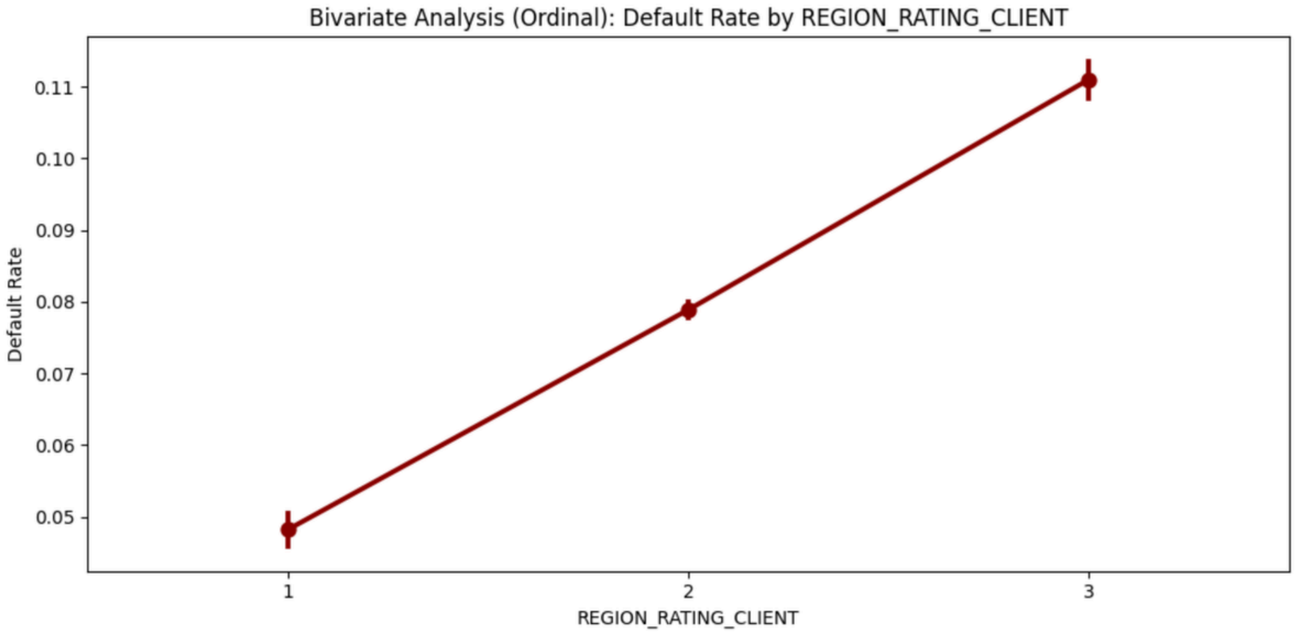
```
Nuclear Stacking Ensemble evaluation complete. RMSE: 22.3446
```

- Ensemble models outperform linear baselines
- XGBoost improvement over Random Forest is statistically significant ($p = 0.026$)
- But the overall ceiling is modest, as the best RMSE of 22.3 on a 0–100 scale means meaningful prediction error remains

Categorical Variable Analysis

Ordinal variables → Pointplots





Categorical Variable Analysis

Key Takeaways for Ordinal variables

Inverse Relationship between education and default risk

1. **The High-Risk Group:** Applicants who only completed "Lower secondary" education are the riskiest (11% default rate)
2. **The Low- Risk Group:** The line drops onsistently with every degree earned. By the time an applicant reaches "Academic degree," their default risk drops to roughly 2%.
3. **Business Insight**: Investing in highly educated applicants is one of the safest bets the bank can make.

Inverse Relationship between Region Rating and default risk

1. Applicants from region 1 are reliable (~5% default rate), whereas region 3 are risky (~11% default rate)
2. **Business Insight**: Checking regional rating is an important factor.

Heatmap Analysis

Trying to understand how quantitative variables impact Target

METHODOLOGY:

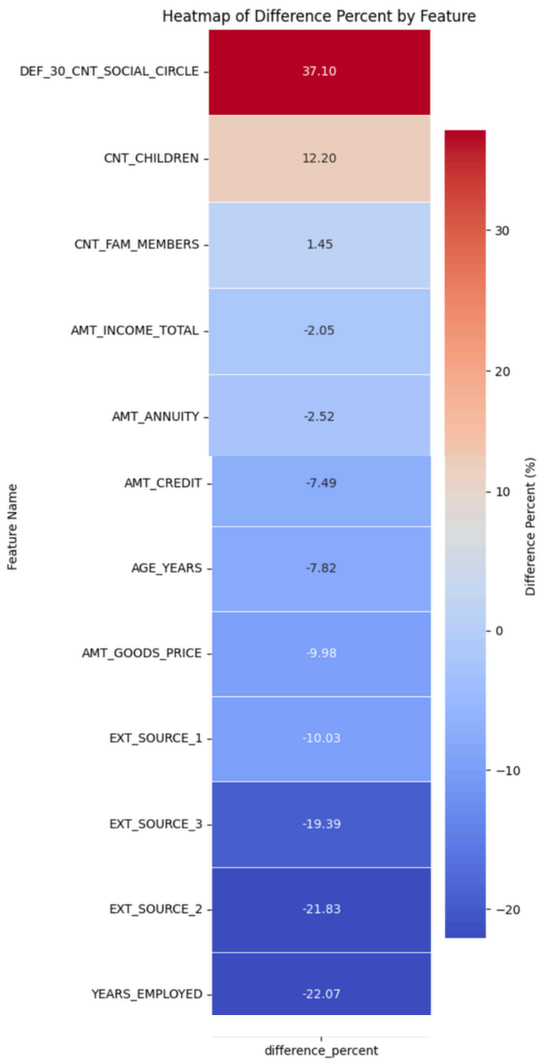
- Separate the data into two groups: TARGET=0 and TARGET=1
- Calculate the mean values for several different quantitative columns for both groups
- Calculate the difference of the means and divide it by the mean for both groups
- Look at the percentage difference for all the variables

Heatmap Analysis

```
TARGET
0    0.508396
1    0.457791
Name: EXT_SOURCE_1, dtype: float64
The DIFFERENCE IS -10.03%

TARGET
0    0.52357
1    0.41126
Name: EXT_SOURCE_2, dtype: float64
The DIFFERENCE IS -21.83%

TARGET
0    44.214182
1    40.780351
Name: AGE_YEARS, dtype: float64
The DIFFERENCE IS -7.82%

TARGET
0    0.523767
1    0.423775
Name: EXT_SOURCE_3, dtype: float64
The DIFFERENCE IS -19.39%

TARGET
0    0.138663
1    0.191702
Name: DEF_30_CNT_SOCIAL_CIRCLE, dtype: float64
The DIFFERENCE IS 37.10%
```

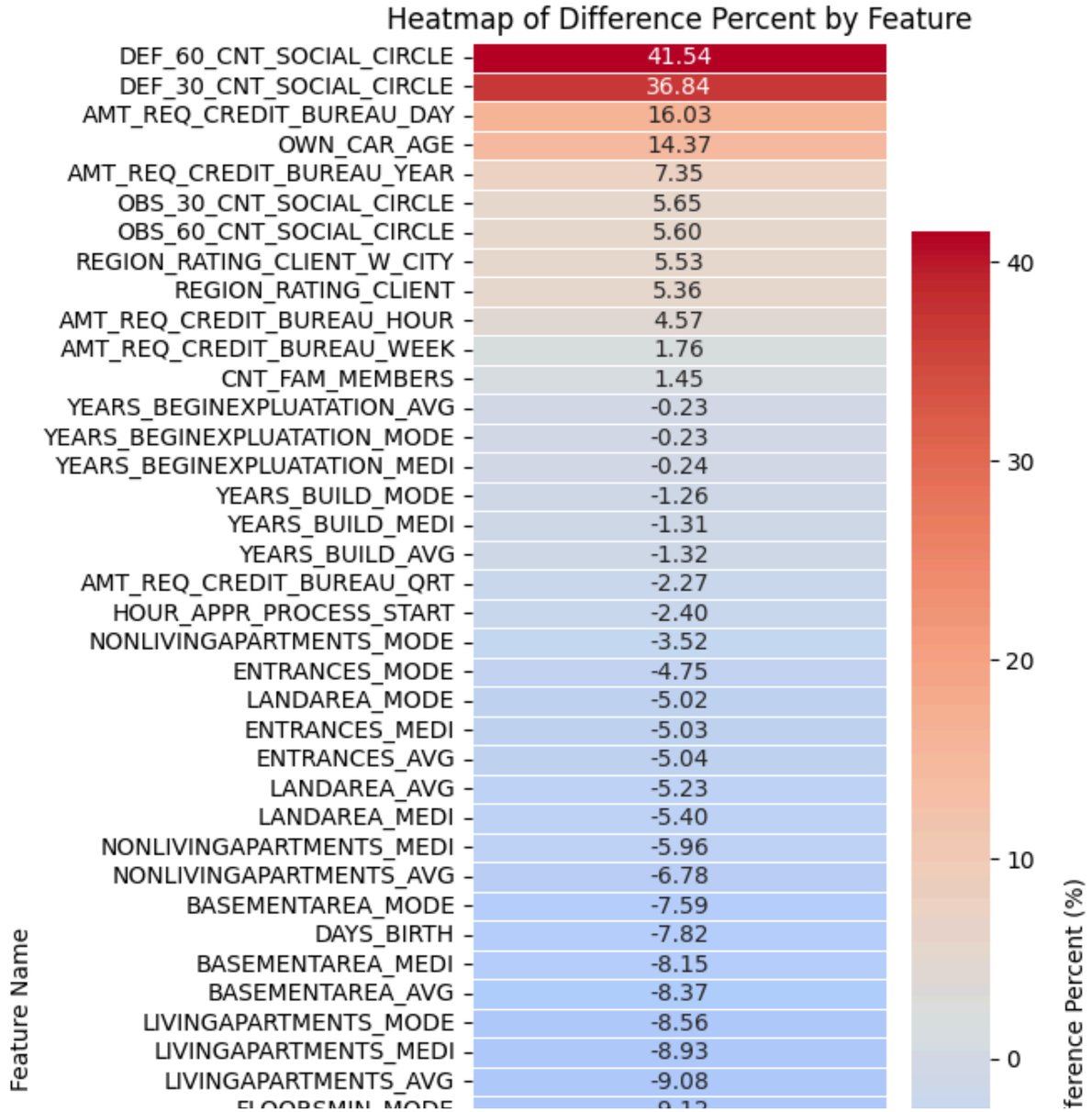
```
TARGET
0    0.412946
1    0.463807
Name: CNT_CHILDREN, dtype: float64
The DIFFERENCE IS 12.20%

TARGET
0    2.150152
1    2.181269
Name: CNT_FAM_MEMBERS, dtype: float64
The DIFFERENCE IS 1.45%

TARGET
0    27163.527386
1    26481.744290
Name: AMT_ANNUITY, dtype: float64
The DIFFERENCE IS -2.52%

TARGET
0    542652.484651
1    488939.444955
Name: AMT_GOODS_PRICE, dtype: float64
The DIFFERENCE IS -9.98%

TARGET
0    169077.722266
1    165611.760906
Name: AMT_INCOME_TOTAL, dtype: float64
The DIFFERENCE IS -2.05%

TARGET
0    602648.282002
1    557778.527674
Name: AMT_CREDIT, dtype: float64
The DIFFERENCE IS -7.49%

TARGET
0    6.278687
1    4.917300
Name: YEARS_EMPLOYED, dtype: float64
The DIFFERENCE IS -22.07%
```

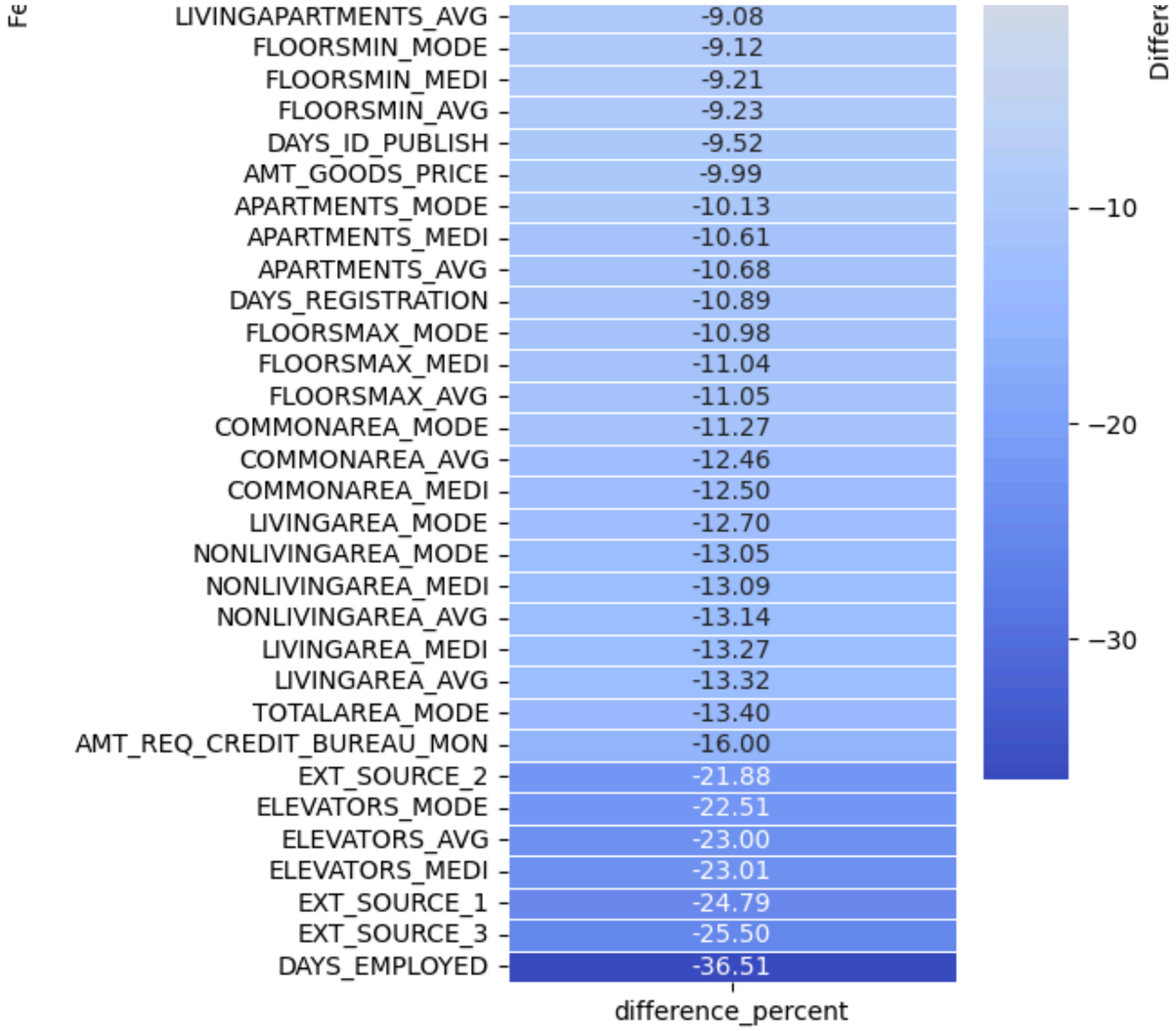
Heatmap Analysis <u>for selected variables</u>



Heatmap Analysis

for entire numeric variables





Heatmap Analysis

A high positive difference means that a TARGET value of 1 has a higher average of that variable. This means that the company should be putting a higher risk to a high number for that variable.

Strong Positive Risk Signals (Defaulters have much higher values)

DEF_60_CNT_SOCIAL_CIRCLE (+41.54%)
DEF_30_CNT_SOCIAL_CIRCLE (+36.84%)
AMT_REQ_CREDIT_BUREAU_DAY (+16%)
OWN_CAR_AGE (+14%)
REGION_RATING_CLIENT (+5%)
OBS_30_CNT_SOCIAL_CIRCLE (+5%)

Interpretations:

- Social circles are an important default interpreter.
- Applicants with other payment difficulties are more likely to default.
- Applicants who default have older cars on average.

Heatmap Analysis

A high positive difference means that a TARGET value of 1 has a higher average of that variable. This means that the company should be putting a higher risk to a high number for that variable.

Strong Negative Risk Signals (Defaulters have much lower values)

DAYS_EMPLOYED (–36.51%)

EXT_SOURCE_3 (–25.5%)

EXT_SOURCE_1 (–24.79%)

EXT_SOURCE_2 (–21.88%)

AMT_REQ_CREDIT_BUREAU_MON (–16%)

LIVINGAREA variables (–10 to –13%)

Interpretations:

- Defaulter had been working for a lower amount of time.
- Defaulters have smaller houses.
-

Historical Data Analysis

Credit Ratio Analysis

Methodology

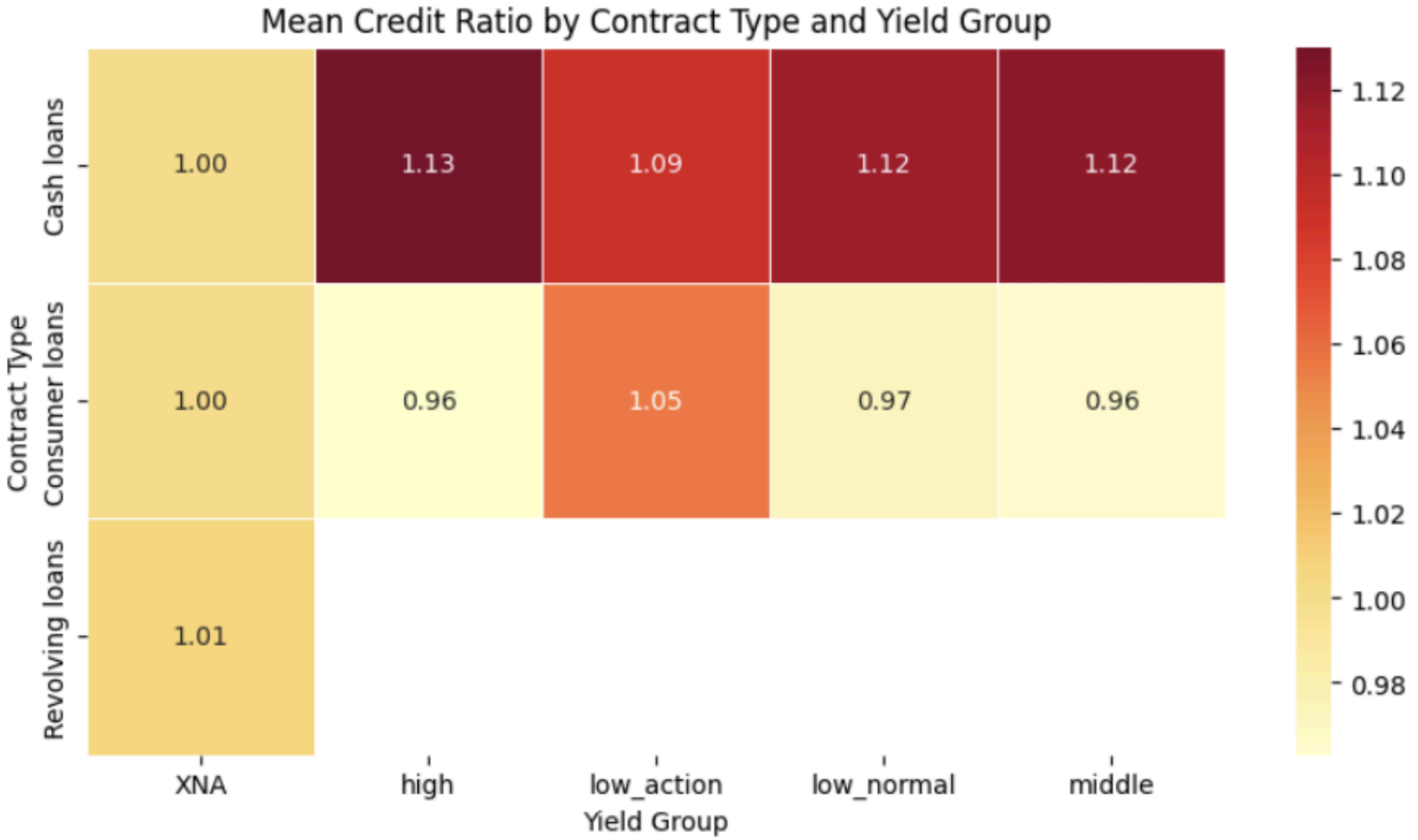
- Calculated a new column, CREDIT_RATIO by dividing AMT_CREDIT by AMT_APPLICATION to measure how much credit was granted relative to what was applied for
- Filtered out unreliable data by keeping only rows where CREDIT_RATIO was less than 2, removing division by zero errors and outliers
- Grouped the data by NAME_CONTRACT_TYPE and NAME_YIELD_GROUP and calculated the mean credit ratio for each combination
- Unstacked the result into a pivot table with contract types as rows and yield groups as columns
- Visualised the pivot table as a diverging heatmap centered at 1.0, where red indicates the bank granted more than applied for and yellow indicates less

New Variable

CREDIT_RATIO → LOGIC → AMT_CREDIT / AMT_APPLICATION

Historical Data Analysis

Credit Ratio Analysis



Historical Data Analysis

Credit Ratio Analysis Key Takeaways

Cash loans consistently receive more credit than requested

1. **Risk Signal**: Cash loan applicants receive 9–13% more credit than they apply for across all yield groups, suggesting the bank is actively upselling credit
2. **Business Insight**: Approving more credit than requested increases the repayment burden on the applicant, which could be a contributing factor to default risk

Consumer loans are more likely to receive less than applied for

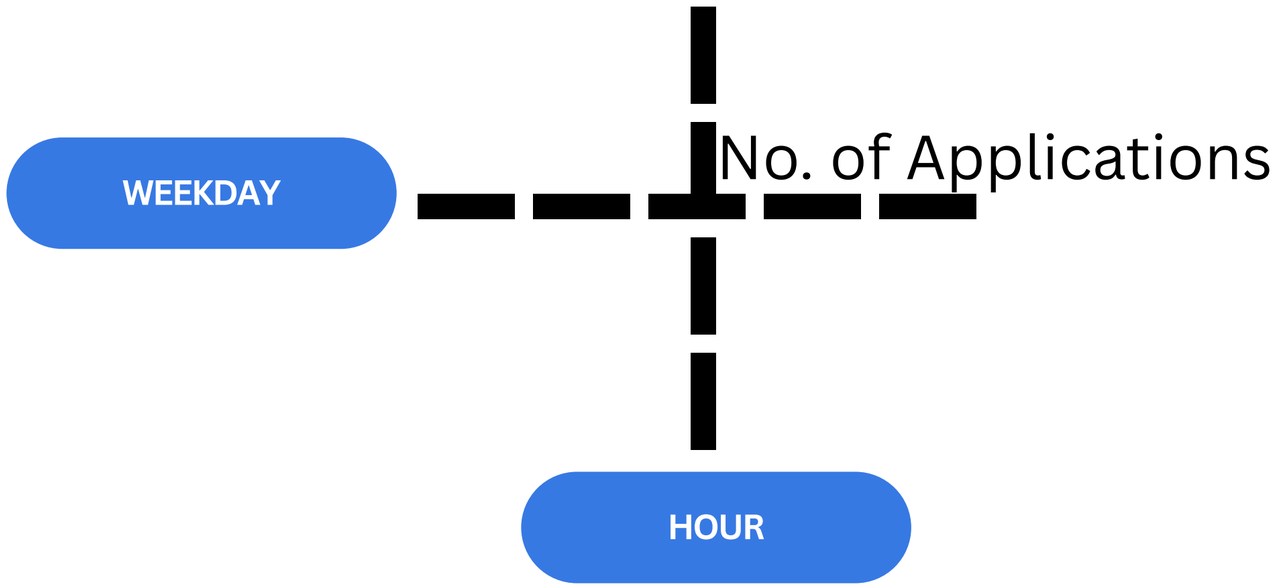
1. **Risk Signal**: Consumer loan ratios fall below 1.0 for most yield groups, meaning the bank is already being cautious, likely because this segment has a higher observed default rate
2. **Business Insight**: The bank's conservative approach to consumer loans may be justified, and yield group should be used as an active filter when evaluating consumer loan applications

Historical Data Analysis

Temporal Analysis of Application Volume

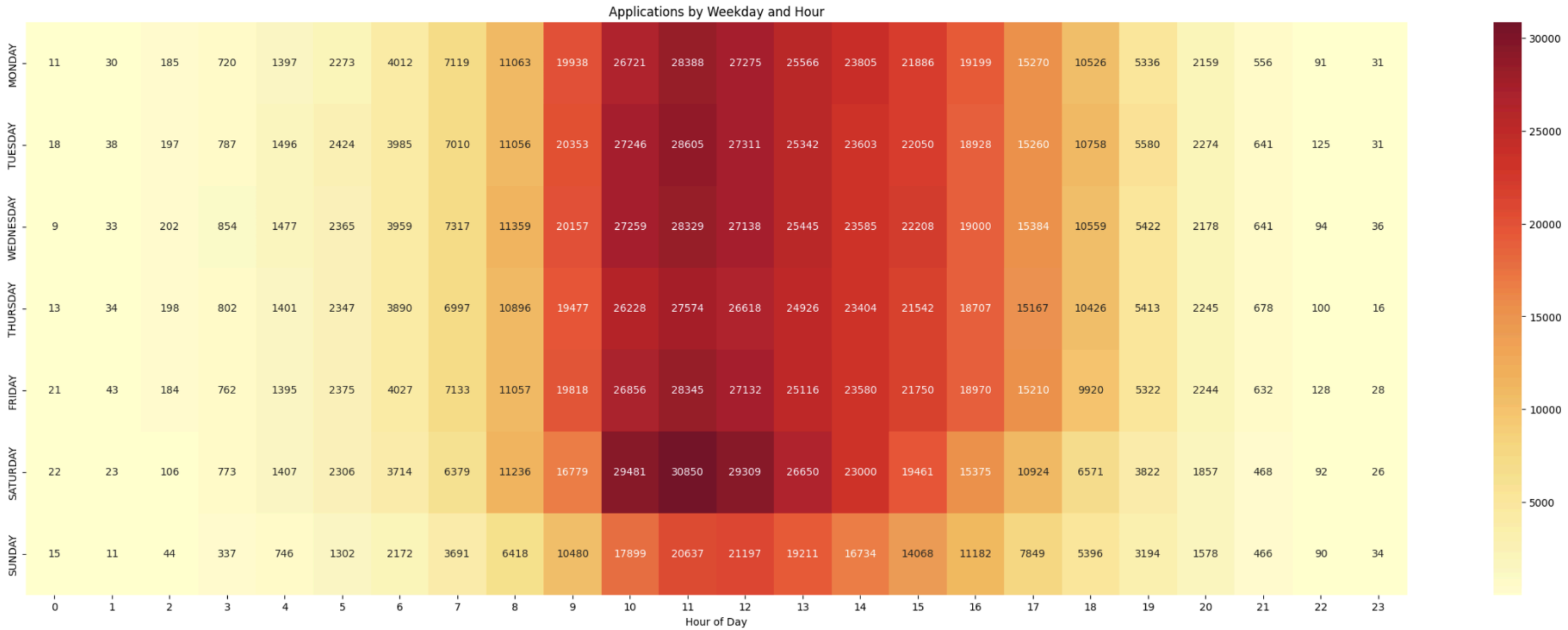
Methodology

- Grouped the data by WEEKDAY_APPR_PROCESS_START and HOUR_APPR_PROCESS_START and counted the number of applications for each combination
- Unstacked the result into a pivot table and reindexed rows to follow Monday–Sunday order
- Visualised as a heatmap where darker red indicates higher application volume



Historical Data Analysis

Temporal Analysis of Application Volume



Historical Data Analysis

Temporal Analysis of Application Volume Key Takeaways

Peak hours are consistent across all weekdays

1. **The Busiest Window**: Applications surge between 10am–12pm every day, with Saturday hitting the single highest count of 30,850 at 11am
2. **Business Insight**: Banks should ensure maximum staffing and system capacity during the 10am–12pm window across all days

Peak application hours are concentrated in the late morning

1. **The Risk Signal**: Saturday at 11am sees the highest volume (30,850 applications), and weekends overall show strong activity — these may be periods of less thorough review
2. **Business Insight**: High-volume periods with potentially stretched review capacity could lead to riskier loans being approved — flagging applications submitted during peak hours for additional scrutiny could reduce default rates

Thank You

- Mauurya
- Mayank